

ABERDARE TECHNICAL COLLEGE
INFORMATION COMMUNICATION TECHNOLOGY

MODULE 2, TERM 2

PRACTICAL EXAMINATION 1 – OBSERVATION CHECKLIST

UNIT OF COMPETENCY: NETWORK SETUP

UNIT CODE: 0612 351 03A

LEVEL: 5

ELEMENT 1: TERMINATE NETWORK CABLES

TIME: 2 HOURS

CANDIDATE DETAILS

Candidate Name: _____

Admission Number: _____

Assessor Name: _____

Date: _____

Venue: _____

Signature: _____

CUTTING LIST

No.	Item	Quantity	Provided By
1	UTP Cat 5e/Cat 6 network cable	2 lengths	Assessor
2	RJ45 connectors	6 pieces	Assessor
3	RJ45 crimping tool	1	Assessor
4	Cable stripper	1	Assessor

5	Cable tester	1	Assessor
6	Cable labels or marker pen	1	Assessor

A. PRACTICAL OBSERVATION CHECKLIST – 50 MARKS

No.	Items to be Evaluated	Allocated Marks	Candidate Score	Comments
<i>TASK 1: TERMINATE A STRAIGHT-THROUGH CABLE</i>				
1.1	Selects the required UTP cable, RJ45 connectors and termination tools	3		
1.2	Measures and cuts the UTP cable appropriately	3		
1.3	Removes the outer jacket without damaging the conductors	4		
1.4	Separates and arranges the conductors correctly	3		
1.5	Arranges the conductors according to the T568B colour sequence	5		
1.6	Inserts the conductors fully and correctly into the RJ45 connector	4		
1.7	Uses the crimping tool correctly to secure the RJ45 connector	3		
1.8	Terminates the second end using the T568B standard	3		
<i>TASK 2: TERMINATE A CROSSOVER CABLE</i>				
2.1	Prepares the UTP cable and RJ45 connectors	2		

2.2	Removes the outer jacket without damaging the conductors	3		
2.3	Arranges the conductors according to the T568A colour sequence	4		
2.4	Inserts the conductors correctly into the RJ45 connector	3		
2.5	Uses the crimping tool correctly to secure the connector	2		
2.6	Arranges the conductors according to the T568B colour sequence on the opposite end	4		
2.7	Completes the crossover cable termination correctly	2		
<i>TASK 3: TEST AND PRESENT THE CABLES</i>				
3.1	Connects the straight-through cable correctly to the cable tester	1		
3.2	Observes and records the tester sequence	1		
TOTAL PRACTICAL		50		

B. PRODUCT CHECKLIST – 10 MARKS

Note: Product assessment is separate from the 50 practical marks.

No.	Product Items to be Evaluated	Allocated Marks	Candidate Score	Comments
1	Straight-through cable is neatly terminated using T568B on both ends	3		
2	Straight-through cable gives the correct tester result	2		

3	Crossover cable is correctly terminated using T568A on one end and T568B on the other	3		
4	Crossover cable gives the correct tester result	1		
5	Both cables are clearly labelled and neatly presented	1		
TOTAL PRODUCT		10		

C. ORAL CHECKLIST – 30 MARKS

No.	Oral Question	Expected Answer	Allocated Marks	Candidate Score	Comments
1	What is a straight-through cable?	A network cable terminated using the same wiring standard at both ends.	3		
2	Which wiring standard is used on both ends of the straight-through cable in this practical?	T568B on both ends.	2		
3	State the T568B colour sequence from pin 1 to pin 8.	White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown.	4		
4	What is a crossover cable?	A network cable terminated using T568A at one end and T568B at the other end.	3		

5	Which wiring standards are used to terminate a crossover cable?	T568A on one end and T568B on the other end.	3		
6	State the T568A colour sequence from pin 1 to pin 8.	White/Green, Green, White/Orange, Blue, White/Blue, Orange, White/Brown, Brown.	4		
7	What is the purpose of an RJ45 connector?	It terminates the network cable and provides a connection to a network port/device.	2		
8	What is the purpose of a crimping tool?	To secure the RJ45 connector firmly onto the network cable conductors.	2		
9	Why should the conductors reach the end of the RJ45 connector before crimping?	To ensure proper contact between the conductors and connector contacts.	2		
10	What is the purpose of a cable tester?	To check continuity and verify the correct wiring of the network cable.	2		
11	What should you do if the cable tester shows a fault?	Identify the fault, correct or re-terminate the cable, and test it again.	2		
12	Why should the completed cables be labelled?	To identify the type of cable and prevent confusion during use.	1		
TOTAL ORAL			30		

D. OVERALL ASSESSMENT RECORD

Assessment Component	Maximum Marks	Candidate Score
Practical Observation Checklist	50	
Product Checklist	10	
Oral Checklist	30	
TOTAL	90	

COMPETENCE DECISION

Decision	Tick
Competent	
Not Yet Competent	

Assessor's Name: _____

Assessor's Signature: _____